

TECHNICAL REPORT

Title: Performance Analysis of a Solar-Powered Water Pumping System

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1. Abstract

This report analyses the performance, efficiency, and operational behavior of a solar-powered water pumping system installed for agricultural irrigation. Data was collected over four weeks to evaluate solar input, pump output, and overall system reliability. Results show an average system efficiency of 72% and a 35% decrease in electricity consumption.

2. Introduction

Solar-powered pumping systems provide a sustainable alternative to conventional electric pumps. The purpose of this study is to assess the performance of a 1.5 kW solar pump installed in a rural agricultural field and determine its advantages, limitations, and economic benefits.

2.1 Objectives

- Evaluate daily water discharge rate.
- Measure system efficiency under variable solar radiation.
- Analyze cost savings and environmental benefits.

2.2 Scope

The report focuses on technical performance, environmental factors, and basic economic analysis. Maintenance and long-term degradation are excluded.

3. System Description

3.1 Components

- Solar Panels: 8 panels (250 W each), total 2 kW
- Pump Type: DC submersible pump
- Controller: MPPT-based solar pump controller
- Storage: 50,000-liter overhead tank

3.2 Working Principle

Solar panels convert sunlight into electrical energy, which powers the DC pump. The controller ensures optimal power transfer using MPPT technology. Water is pumped into the storage tank for distribution.

4. Methodology

4.1 Data Collection

- Daily solar irradiance (W/m^2)
- Pump discharge rate (liters/hour)
- Energy consumption (kWh)
- Ambient temperature

4.2 Testing Conditions

- Duration: 1 month
- Location: Open agricultural field
- Weather: Mostly sunny with intermittent clouds

5. Results and Analysis

5.1 Solar Energy Input

Average solar irradiance recorded: 820 W/m^2 .

5.2 Pump Output

- Maximum discharge: 18,000 L/day
- Minimum discharge: 11,000 L/day
- Average: 14,500 L/day

5.3 System Efficiency

The overall efficiency ranged from 68% to 75%, averaging 72%.

5.4 Cost Analysis

- Reduction in electricity costs: 35%
- Estimated payback period: 4 years

5.5 Observations

- Efficiency dropped on cloudy days.
- System performed optimally when panel temperature was below 45°C.

6. Discussion

The solar pumping system demonstrated reliable performance with significant energy savings. Seasonal variations affect productivity, but overall efficiency remains high. MPPT controllers improve energy extraction even in low-light conditions.

7. Conclusion

The solar-powered water pump is an effective and eco-friendly alternative for irrigation. It offers high efficiency, low operational cost, and minimal maintenance requirements. Adoption of such systems can help farmers reduce dependency on grid electricity.

8. Recommendations

- Install panel cleaning mechanisms to maintain efficiency.
- Use battery storage for cloudy-day operation.
- Conduct long-term performance studies.

9. References

1. Manufacturer's technical datasheet
2. Solar Pumping Guide, Ministry of Renewable Energy
3. Field measurement logs (2025)